INCH-POUND

MIL-PRF-38135D (USAF)
16 December 2014
SUPERSEDING
MIL-PRF-38135C (USAF)
25 November 1997

PERFORMANCE SPECIFICATION

INDICATOR, INDICATED AIRSPEED AVU-19A, STANDBY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the standby, integrally-lighted, 1-7/8 inch, round dial pitot static airspeed indicator, designated AVU-19/A.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Oklahoma City Air Logistics Center/ENSDAA, 3001 Staff Drive, Tinker AFB, OK 73145 or emailed to ocalc.dsp@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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FEDERAL STANDARDS

FED-STD-595/37038	- Black, Lusterless
FED-STD-595/37875	- White, Lusterless

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-14806	- Coating, Reflection Reducing for Instrument Cover Glasses and Lighting Wedges "Inactive for New Design"
MS28105	- Cover Glass, Aircraft Instrument Dial
MS33558	- Numeral and Letter, Aircraft Instrument Dial, Standard Form of
MS33638	- Cases, Instrument, Flange Mounted, Aircraft "Inactive for New Design"

(Copies of these documents are available online at http://quicksearch.dla.mil/).

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ-Z1.4	- Sampling Procedures and Tables for Inspection by Attributes

(ASQ documents may be obtained at http://asq.org or addressed to the American Society for Quality, P.O. Box 3005, Milwaukee, WI 53201-3005 or 600 North Plankinton Avenue, Milwaukee, Wisconsin, 53203, USA)

ASTM INTERNATIONAL

ASTM-B117	- Salt Spray (Fog) Apparatus, Operating

(ASTM documents may be obtained online at http://www.astm.org or from American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

RADIO TECHNICAL COMMISSION FOR AERONAUTICS (RTCA)

RTCA/DO-160	- Environmental Conditions and Test Procedures for Airborne Equipment

(RTCA documents may be obtained online at www.rtca.org or addressed to RTCA Inc, 1828 L Street, NW, Suite 805, Washington, DC 20036.)

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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE-AS5202 - Port or Fitting End, Internal Straight Thread

(SAE documents may be obtained at http://www.sae.org/ or from Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials. All materials shall be suitably treated to resist corrosion due to electrolytic decomposition, fungus, salt spray, and any other atmospheric condition that may be encountered during operational use or storage. Non-magnetic materials shall be used, where feasible. The use of toxic chemicals, hazardous substances, or ozone depleting chemicals shall be avoided, whenever feasible.

3.4 Interface.

3.4.1 Case. The case shall conform to MS33638 for the 2-inch nominal size except as shown on figure 1. The bezel and that part of the case visible from the cockpit side of the instrument panel shall be finished in lusterless black, color number 37038 of FED-STD-595. The pitot and static pressure bosses shall be marked with a P and an S as shown in figure 1.

3.4.2 Cover glass. The cover glass shall conform to MS28105 and shall be sealed into the bezel.

3.4.3 Reflection reducing coating. The cover glass surface shall be coated to reduce reflection according to MIL-C-14806.

3.4.4 Electrical connector. The electrical connector shall conform to figure 1.

3.4.5 Dial markings. All dial markings shall be permanently and legibly marked and shall conform to figure 2 and table I. All letters and numbers shall conform to MS33558. All

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markings not otherwise specified shall be finished in lusterless black, color number 37038 of FED-STD-595.

TABLE I. Dial markings.

Marking	Height or Length	Width	Color (see note 3)
Numerals	0.140	0.020	lusterless white, 37875
100-knot graduations	0.171	0.020	lusterless white, 37875
50-knot graduations	0.156	0.020	lusterless white, 37875
10-knot graduations	0.140	0.015	lusterless white, 37875
USAF	0.047	0.020	lusterless black, 37038
AVU-19/A	0.047	0.020	lusterless black, 37038
AIRSPEED	0.047	0.020	lusterless black, 37038
KNOTS X 100	0093	0.020	lusterless black, 37038

NOTES: 1. All dimensions are in inches.
2. Tolerances: height or length $\pm$0.016; width $\pm$0.005.
3. All color numbers are according to FED-STD-595.
4. Height refers to characters; length refers to graduations.
5. USAF, AVU-19/A, and AIRSPEED shall be in raised lettering.

3.5 Performance.

3.5.1 Operation. The indicator shall display the indicated airspeed (IAS) as a function of the difference between pitot and static pressures.

3.5.2 Maintainability. The indicator shall be repairable without the use of special tools.

3.5.3 Reliability. The indicator shall have a mean time between failures (MTBF) of 1,200 hours.

3.5.4 Range. The indicator shall display the IAS over a range of 60 to 850 knots. The pointer shall move continuously (without ratcheting) while indicating IAS.

3.5.5 Overpressure. The indicator shall withstand overpressure of 10 inches of water at the static port and 73 inches of Hg at the pitot port.

3.5.6 Magnetic effect. The indicator shall cause a free compass to deflect no more than 1° when revolved about a short-bar magnetic compass with a field intensity of 0.17 to 0.19 oersted.

3.5.7 Case leakage. The case leakage shall cause a change of pressure of no more than 0.01 inch Hg per minute.

3.5.8 Static pressure. The pointer shall not change position due to internal leakage more than 1 knot when a pressure sufficient to cause full-scale deflection is applied.

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3.5.9 Scale error. The scale error (see 6.3.1) of the indicator shall be within the tolerances specified in table II.

TABLE II. Scale error tolerances.

Test point	Differential pressure (inches Hg)	Tolerance (±knots)
0	0	10
60	0.172	3
80	0.308	3
100	0.482	3
120	0.696	3
150	1.092	3
200	1.962	5
300	4.541	10
400	8.397	15
500	13.797	20
600	21.109	20
700	30.816	20
200	43.009	20

3.5.10 Friction error. The friction error shall be within the tolerances in table III.

3.5.11 Position error. The position error shall be within the tolerances in table III.

TABLE III. Friction and position error tolerances.

Indicated airspeed (knots)	Tolerance (±knots)
60 to 200	3
201 to 850	10

3.5.12 Dielectric strength. The indicator shall withstand application of 500Vac between the pins and case for 5 seconds without insulation breakdown.

3.5.13 Dial and cover glass location. The distance between the front surface of the bezel and the outside of the cover glass shall be 0.062 inch or less. The distance between the inside surface of the cover glass and the dial shall be 0.125 inch or less.

3.5.14 Pointer. There shall be no oscillation in excess of ±5 knots of the pointer while the indicator is vibrated. The pointer shall be adjustable. The pointer shall be finished in lusterless white, color number 37875 of FED-STD-595.

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3.5.15 Limitation of pointer movement. The airspeed pointer movement shall be limited to 350° of clockwise rotation from the no pressure position.

3.5.16 Lighting characteristics. The lighting characteristics (intensity and color) shall be as specified by the procuring activity (see 6.2), and shall operate on 5Vac.

3.5.17 Damping. The time for the pointer to go from full scale to 120 knots shall be between 0.8 and 2.0 seconds (see 6.4).

3.5.18 Weight. The indicator shall weigh 1.25 pounds or less.

3.5.19 Dial. The dial shall conform to figure 2.

3.6 Item identification. The indicator shall be permanently and legibly marked with the following information:

a. Manufacturer's name.
b. Manufacturer's CAGE code.
c. Manufacturer's part number.
d. Manufacturer's lot number.
e. National Stock Number (NSN).
f. Date of manufacture.

3.7 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.8 Environmental conditions.

3.8.1 Temperature. The indicator shall operate from -65° to 160°F with a scale error of $\pm$5 knots in addition to the scale error in table II.

3.8.2 Vibration.

3.8.2.1 Vibration error. The indicator shall withstand vibration at an amplitude of 0.003 to 0.005 inch at a frequency of 500 to 3,000 Hz with an error less than 5 knots at an IAS of 200 knots.

3.8.2.2 Vibration tolerance. The indicator shall withstand vibration at an amplitude of 0.018 to 0.020 inch at frequencies of 1,000, 2,000, and 3,000 Hz for a period of 3 hours with an error less than 5 knots at an IAS of 200 knots.

3.8.2.3 Resonance. The indicator shall have no resonant frequency between 500 and 3,000 Hz.

3.8.3 Humidity. The indicator shall operate in humidity up to 95 percent.

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3.8.4 Acceleration. The indicator shall operate with applied acceleration forces up to 20g.

3.8.5 Sand and dust. The indicator shall operate after exposure to sand and dust.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When required (see 6.2), first article inspection shall be performed on nine indicators. Three indicators shall be subjected to all tests except reliability. The remaining six indicators shall only be subjected to the reliability test.

4.3 Conformance inspection. Conformance inspection shall consist of the following:

a. Individual tests (see 4.3.1)
b. Sampling tests (see 4.3.2).

4.3.1 Individual tests. Each indicator shall be subjected to the following tests:

a. Examination (see 4.6.1).
b. Scale error (see 4.6.4)
c. Friction error (see 4.6.5).
d. Case leakage (see 4.6.10).
e. Static pressure test (see 4.6.11).
f. Lighting (see 4.6.14).

4.3.2 Sampling tests. Indicators, sampled according to ANSI/ASQ-Z1.4, shall be subjected to the following tests:

a. Overpressure (see 4.6.8).
b. Magnetic effect (see 4.6.9).
c. Case leakage (see 4.6.10).
d. Dielectric strength (see 4.6.12).
e. Pointer oscillation (see 4.6.13).
f. Damping (see 4.6.15).
g. Low temperature (see 4.6.16).
h. High temperature (see 4.6.17).
i. Vibration error (see 4.9.18).
j. Vibration tolerance (see 4.6.19).
k. Pointer zero position error (see 4.6.23).

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4.4 Test conditions. Unless otherwise specified (see 6.2), all tests shall be performed in accordance with the test conditions specified in the applicable test method document or applicable paragraph in this specification.

4.4.1 Standard atmospheric condition. Unless otherwise specified in this specification or the applicable test method document, all tests shall be performed at ambient pressure altitude and temperature.

4.4.2 Tapping. Unless otherwise specified in this specification, the indicator shall be tapped before a test reading is taken.

4.4.3 Attitude. Unless otherwise specified in this specification or the applicable test method document, the indicator shall be tested in the normal operating position.

4.5 Requirements cross reference matrix. Table IV provides a cross reference matrix of the section 3 requirements tested or verified in the paragraphs below.

TABLE IV. Requirements cross-reference matrix.

Requirement	Verification	Requirement	Verification
3.1	4.2	3.5.12	4.6.12
3.3	4.6.1, 4.6.2, 4.6.3	3.5.13	4.6.1
3.4.1	4.6.1	3.5.14	4.6.1, 4.6.13
3.4.2	4.6.1, 4.6.10	3.5.15	4.6.1, 4.6.11
3.4.3	4.6.24	3.5.16	4.6.14
3.4.4	4.6.1	3.5.17	4.6.15
3.4.5	4.6.1	3.5.18	4.6.1
3.5.1	4.6.4, 4.6.5, 4.6.6	3.5.19	4.6.1
3.5.2	4.6.1	3.6	4.6.1
3.5.3	4.6.7	3.7	4.6.1
3.5.4	4.6.4, 4.6.5, 4.6.6	3.8.1	4.6.16, 4.6.17
3.5.5	4.6.8	3.8.2.1	4.6.18
3.5.6	4.6.9	3.8.2.2	4.6.19
3.5.7	4.6.10	3.8.2.3	4.6.18

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TABLE IV. Requirements cross-reference matrix - (Continued).

Requirement	Verification	Requirement	Verification
3.5.8	4.6.11	3.8.3	4.6.20
3.5.9	4.6.4	3.8.4	4.6.21
3.5.10	4.6.5	3.8.5	4.6.22
3.5.11	4.6.6		

4.6 Tests.

4.6.1 Examination. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, interface, markings, colors, weight, maintainability, item identification, pointer, limitation of pointer movement, pointer zero position, lighting, interchangeability, and dimensions.

4.6.2 Fungus. The indicator, including applicable external connectors, shall be subjected to the fungus test in RTCA/DO-160. The indicator shall then be subjected to the individual tests.

4.6.3 Salt fog. The indicator shall be subjected to the salt fog test in accordance with ASTM-B117 for 50 hours. The indicator shall then be subjected to the individual tests.

4.6.4 Scale error. The indicator shall be subjected to the pressures specified in table II, first with increasing pressure, then decreasing pressure. For increasing pressure, the pressure shall be brought up to but not exceed the pressure required to give the desired reading and for decreasing pressure, the pressure shall be brought down to but not below the pressure required to give the desired reading. At each test point, the scale errors shall be within the tolerance in table II.

4.6.5 Friction error. The indicator shall be tested for friction at the 60, 100, 150, 200, 300, 400, 500, 600, 700, and 800-knot indications. The pressure shall be increased to bring the pointer to the approximate desired reading and held constant while two readings are taken, the first before the indicator is tapped, the second after the indicator is tapped. The difference between the two readings shall be within the tolerances in table III. The pointer shall move smoothly while the pressure is varied uniformly without vibrating the indicator.

4.6.6 Position error. Power and pressure shall be applied to the indicator. The indicator shall be tested for position error at 60, 100, 150, 200, 300, 400, 500, 600, 700, and 800-knot indications. The pressure shall be increased to bring the indicator to the desired reading in the horizontal position, and then again in the vertical position. The difference in reading at each test point shall be within the tolerance in table III.

4.6.7 Reliability. The manufacturer shall demonstrate an MTBF of 1,200 hours with a confidence of 0.90 and discrimination ratio of 2.

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4.6.8 Overpressure. The indicator shall be subjected to a pressure of 10 inches of water at the static connection and 73 inches of Hg at the pitot connection for 10 minutes. Five minutes after the removal of the pressure, the indicator shall be subjected to the scale error test. The errors at the test points shall be less than $\pm$5 knots.

4.6.9 Magnetic effect. The indicator shall be held in various positions with its nearest part five inches from the center of a short-bar magnetic compass with any compensator magnets removed. The compass shall be in uniform magnetic field having a horizontal intensity between 0.17 and0.19 oersted. The maximum deflection shall not exceed 1 degree.

4.6.10 Case leakage. The pitot and static pressure connections of the indicator shall be connected to the same mercury manometer, or equivalent, and source of vacuum and pressure. A vacuum of 25 inches Hg shall be applied to the pitot and static connections of the indicator. With the source connections closed off at the indicator for 1 minute, the pressure in the case shall change no more than 0.01 inch of Hg. This procedure shall be repeated with a pressure of 10 inches Hg. The mercury level shall change no more than 0.01 inch of Hg per minute.

4.6.11 Static pressure test. With the static connection open to the atmosphere, a pressure sufficient to produce full scale deflection of the pointer shall be applied to the pitot connection of the indicator. The pointer shall not proceed past 358° of rotation. All external connections shall then be sealed. The pointer shall change its position by no more than 1 knot per minute.

4.6.12 Dielectric strength. A potential of 500Vac shall be applied simultaneously between each terminal and the case for 5 seconds. There shall be no insulation breakdown or permanent damage to the indicator. The indicator shall then be subjected to individual tests.

4.6.13 Pointer oscillation. The indicator shall be subjected to the vibration error and vibration tolerance tests. There shall be no change in excess of $\pm$5 knots in the pointer indication due to vibration.

4.6.14 Lighting. The indicator shall be subjected to the lighting tests specified by the procuring activity (see 6.2).

4.6.15 Damping. Pressure shall be applied to the pitot connector to produce full-scale deflection of the airspeed pointer. The pressure shall be released. The time required for the airspeed pointer to reach the 120-knot graduation shall be between 0.8 and 2.0 seconds.

4.6.16 Low temperature. The indicator shall be subjected to -65° $\pm$3°F for 48 hours. At the end of 48 hours and while still at -65° $\pm$3°F, the indicator shall be subjected to the scale error test. The error at the test points shall be no more than $\pm$5 knots in addition to the scale error in table II.

4.6.17 High temperature. The indicator shall be subjected to 160° $\pm$3°F for 24 hours. At the end of 24 hours and while still at 160° $\pm$3°F, the indicator shall be subjected to the scale error

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test. The errors at the test points shall be no more than $\pm$5 knots in addition to the scale error in table II.

4.6.18 Vibration error. Pressure shall be applied to produce a deflection of 200 knots. The indicator shall be mounted to the vibration stand in its normal operating position. The indicator shall be subjected to vibration with an amplitude between 0.003 and 0.005 inch frequencies from 500 to 3,000 Hz. The path of vibration shall be in a circle inclined at a 45° angle. All resonant frequencies of the airspeed indicator shall occur outside of this range. While the indicator is being vibrated, the pointer error shall be no more than $\pm$5 knots.

4.6.19 Vibration tolerance. Increasing pressure shall be applied to produce a deflection of 200 knots. The indicator shall be vibrated at 1,000 Hz for 1 hour, followed by 1 hour at 2,000 Hz, and then by 1 hour at 3,000 Hz. The vibration shall be in a circle with a diameter between 0.018 and 0.020 inch. At the end of the vibration period, pressure shall be released. The same increasing differential pressure shall be reapplied. The difference between this indication and that determined prior to the vibration shall be less than $\pm$5 knots.

4.6.20 Humidity. The indicator shall be subjected to the class B humidity test in RTCA/DO-160. The indicator shall then be subjected to the individual tests.

4.6.21 Acceleration. The indicator shall be subjected to an acceleration of 20g for a period of 30 seconds in each of three mutually perpendicular directions. The indicator shall then be subjected to the individual tests.

4.6.22 Sand and dust. The indicator shall be tested according to the sand and dust test in RTCA/DO-160. The indicator shall then be subjected to the individual tests.

4.6.23 Pointer zero position error. NO differential pressure shall be applied to the indicator. The pointer shall be within 10 knots of the zero graduation.

4.6.24 Reflection reducing coating. The reflection reducing coating requirement shall be verified according to the appropriate test method in MIL-C-14806.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The AVU-19A indicator covered by this specification is intended for use in aircraft as a standby instrument to indicate the speed relative to air at sea level and the equivalent sea level speed corresponding to the actual sustaining force of the air at any other altitude.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of ASSIST to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
c. If first article inspection is required (see 3.1 and 4.2).
d. Lighting intensity and color requirements (see 3.5.16).
e. If the test conditions are to be different than specified (see 4.4).
f. Lighting test descriptions (see 4.6.14)
g. Data required.
h. Packaging requirements (see 5.1).

6.3 Definition.

6.3.1 Scale error. Scale error is the difference between the indication of the instrument and the calibrated airspeed corresponding to the applied differential pressure.

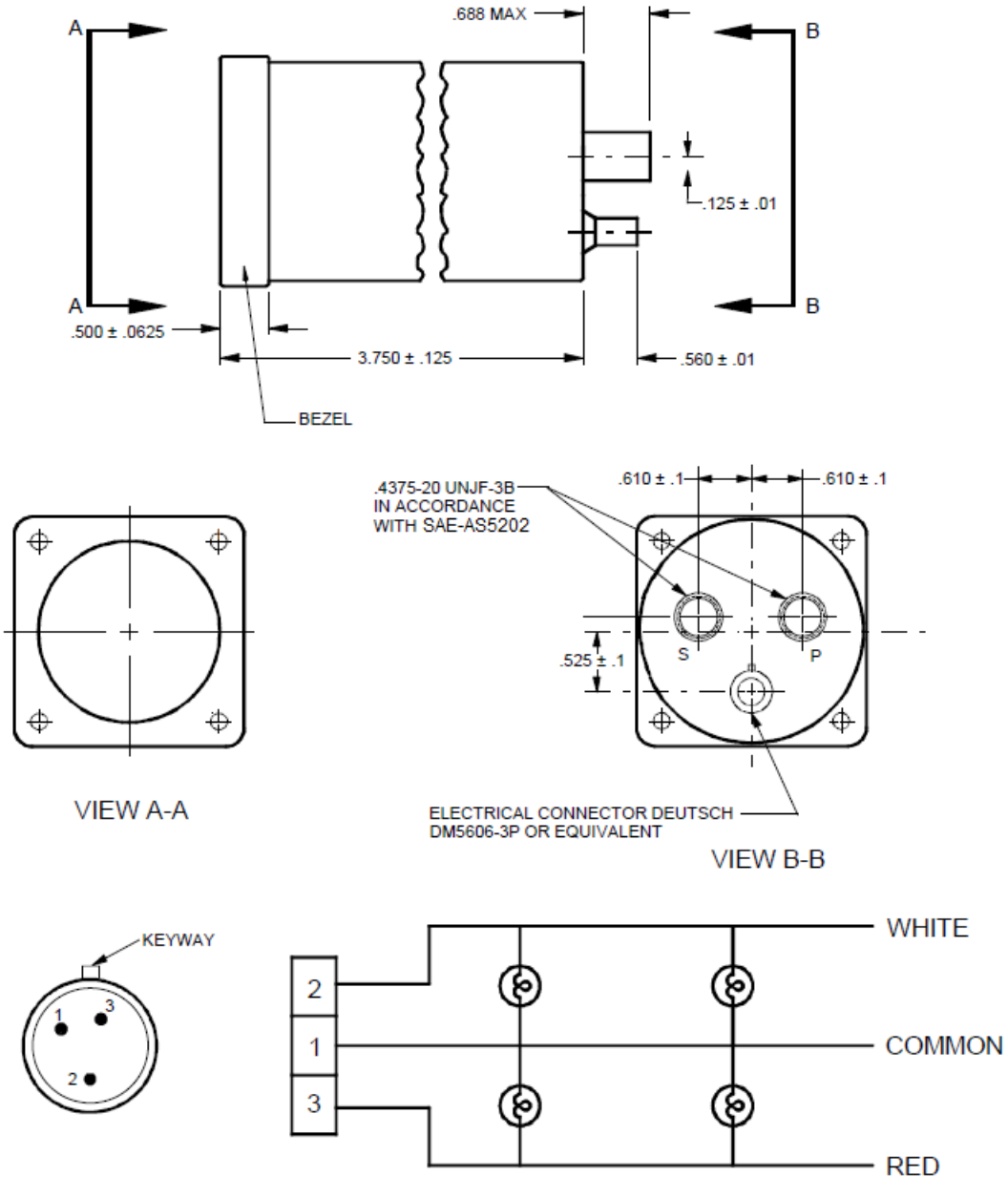
6.4 Damping. Previous indicators were equipped with a length of capillary tubing having a minimum inside diameter of 0.02 inch between the pitot connection in the rear of the case and the diaphragm capsule. The effect of this restriction was to ensure the indicator met damping requirements.

6.5 Subject term (key word) listing.

Backup
Pitot
Static

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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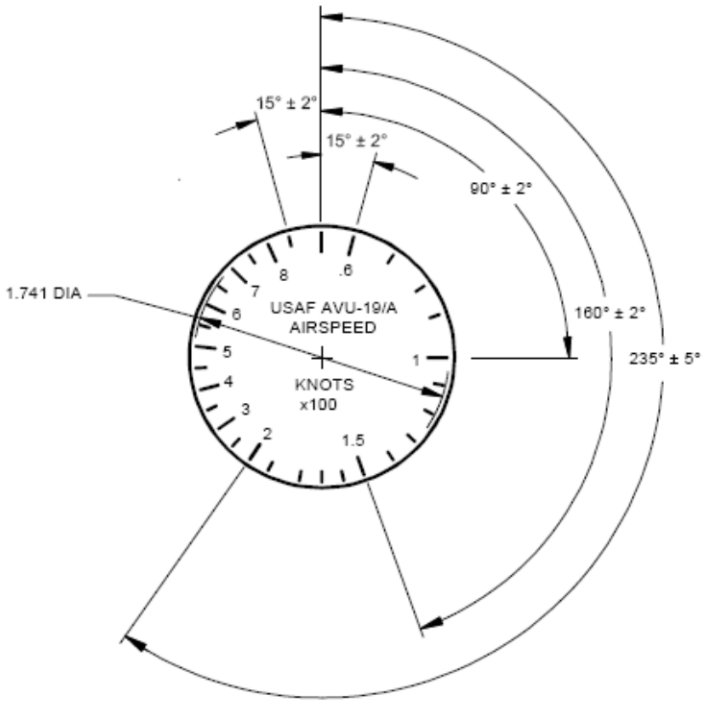


NOTES:

1. DIMENSIONS IN INCHES.
2. DRAWING NOT TO SCALE.
3. THE FLANGE SHALL BE IN ACCORDANCE WITH MS33638.

FIGURE 1. Case.

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NOTES:

1. UNLESS OTHERWISE SPECIFIED, DIMENSIONS IN INCHES.
2. TOLERANCES: DECIMALS ±0.010
3. THE FACE SHALL BE LUSTERLESS BLACK, COLOR NUMBER 37038 OF FED-STD-595.

FIGURE 2. Dial.

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